Syllabus Details

Syllabus ID:	13801
Syllabus Name:	**Integrate single page application with Spring Boot_Tích hợp ứng dụng trang đơn với Spring Boot**
Course Name English:	**Integrate single page application with Spring Boot**
Subject Code:	**SBA301**
Learning-Teaching Method:	Project based learning,Problem based learning,Active learning,In-class lecture
NoCredit:	6
Degree Level:	Bachelor
Time Allocation:	Study hour (300h) = 90h contact hours + 145-minute final exam + 207.6h self-study
Pre-Requisite:	HSF302
Description:	Upon completion of this course students shoud: 1. Understand the followings: • ReactJS concepts for building Front-End application • RESTful API concepts • RESTful API using Spring Boot's built-in features and frameworks • Security features (authentication and authorization) with RESTful API • Handling HTTP requests and responses using Spring Boot's framework • Error handling and exception management in Spring Boot RESTful APIs 2. Be able to: • Build RESTful APIs using Spring Boot's built-in features and frameworks • Implement security features with authentication and authorization • Handle HTTP requests and responses using Spring Boot's framework • Build Front-End application using ReactJS components • Integrate Spring Boot with other technologies and frameworks SQL Server, MongoDB, and ReactJS 3. Be able to work in team and present group's results
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU CMS at http://flm.fpt.edu.vn for up-to-date course information
Tools:	- Internet - IntelliJ IDEA - JDK 21 - Platform: SpringBoot 3, Node 21
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1402/QĐ-ĐHFPT dated 12/12/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/12/2025

6 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Spring REST		Spring Academy				True	False	True	https://docs.spring.io/spring-restdocs/docs/current/reference/htmlsingle/ https://spring.io/guides/tutorials/rest/ https://spring.io/guides/gs/rest-service/

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
2	React						True	False	True	https://react.dev/reference/react-dom/components https://react.dev/reference/react/components
3	Master Java Web Services and REST API with Spring Boot		Udemy				False	False	True	https://www.udemy.com/course/spring-web-services-tutorial/
4	React - The Complete Guide 2023		Udemy				False	False	True	https://www.udemy.com/course/react-the-complete-guide-incl-redux/
5	Spring REST: Building Java Microservices and Cloud Applications		Apress	2022	2nd Edition	978-1484274767	False	True	False	
6	Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3		Apress	2022	1st Edition	978-1484287637	False	True	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand of the ReactJS library, services, and dependency injection.
2	CLO2	Apply fetch and catching data in Client-Server Communication
3	CLO3	Describe RESTful APIs using Spring Boot's built-in features and frameworks
4	CLO4	Describe the connecting RESTful Web Service with JPA
5	CLO5	Apply security features such as authentication and authorization
6	CLO6	Create application with Hibernate and Spring Framework.
7	CLO7	Apply Spring Data with MongoDB to build an application.
8	CLO8	Develop teamwork skills, communication skills (including presentation skills)

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction	Offline		IT		SBA301		
2	Working Environment (JDK, Spring Boot, SQL Server, MongoDB, ReactJS)	Offline			IDE: https://spring.io/tools		- Setup Environment	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Group Project Introduction	Offline					- Choose Group - Choose Topic - Collect requirements about topic	
4	Chapter 01. Introduction to ReactJS	Offline	CLO1	ITU	https://react.dev/reference/react https://www.w3schools.com/react/default.asp		- Read the slide, related reference documents - Do exercises - Self practice	
5	Chapter 01. Introduction to ReactJS (contd.)	Offline	CLO1	TU	https://react.dev/reference/react https://www.w3schools.com/react/default.asp		- Read the slide, related reference documents - Do exercises - Self practice	
6	Chapter 01. Introduction to ReactJS (contd.)	Offline	CLO1	TU	https://react.dev/reference/react https://www.w3schools.com/react/default.asp		- Read the slide, related reference documents - Do exercises - Self practice	
7	Chapter 02. React Component	Offline	CLO1	ITU	https://react.dev/reference/react-dom/components https://react.dev/reference/react/components		- Read the slide, related reference document - Do exercises - Self practice	
8	Chapter 02. React Component (contd.)	Offline	CLO1	TU	https://react.dev/reference/react-dom/components https://react.dev/reference/react/components		- Read the slide, related reference document - Do exercises - Self practice	
9	Chapter 02. React Component (contd.)	Offline	CLO1	TU	https://react.dev/reference/react-dom/components https://react.dev/reference/react/components		- Read the slide, related reference document - Do exercises - Self practice	
10	Chapter 03. Introduction to React-Bootstrap	Offline	CLO1	TU	https://react-bootstrap.netlify.app/docs/getting-started/why-react-bootstrap		- Read the slide, related reference document - Do exercises - Self practice	
11	Chapter 03. Introduction to React-Bootstrap (contd.)	Offline	CLO1	TU	https://react-bootstrap.netlify.app/docs/getting-started/why-react-bootstrap		- Read the slide, related reference document - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	Chapter 03. Introduction to React-Bootstrap (contd.)	Offline	CLO1	TU	https://react-bootstrap.netlify.app/docs/getting-started/why-react-bootstrap		- Read the slide, related reference document - Do exercises - Self practice	
13	Chapter 04. Props, State and Context	Offline	CLO1	TU	https://react.dev/learn/passing-props-to-a-component		- Read the slide, related reference document - Do exercises - Self practice	
14	Chapter 04. Props, State and Context (contd.)	Offline	CLO1	TU	https://react.dev/learn/passing-props-to-a-component		- Read the slide, related reference document - Do exercises - Self practice	
15	Chapter 05. React Hook	Offline	CLO1	TU	https://react.dev/reference/react-dom/hooks https://react.dev/reference/react/hooks		- Read the slide, related reference document - Do exercises - Self practice	
16	Chapter 05. React Hook (contd.)	Offline	CLO1	TU	https://react.dev/reference/react-dom/hooks https://react.dev/reference/react/hooks		- Read the slide, related reference document - Do exercises - Self practice	
17	Lab 01. Working with ReactJS application	Offline	CLO1		Lab 01		- Do Lab 01	
18	Lab 01. Working with ReactJS application (contd.)	Offline	CLO1		Lab 01		- Do Lab 01	
19	Chapter 06. Handling Navigation with Routes	Offline	CLO2	ITU	https://reactrouter.com/en/main/components/navigate		- Read the slide, related reference document - Do exercises - Self practice	
20	Chapter 06. Handling Navigation with Routes (contd.)	Offline	CLO2	TU	https://reactrouter.com/en/main/components/navigate		- Read the slide, related reference document - Do exercises - Self practice	
21	Chapter 07. Client-Server Communication	Offline	CLO2	TU	https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction		- Read the slide, related reference document - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
22	Chapter 07. Client-Server Communication (contd.)	Offline	CLO2	TU	https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction		- Read the slide, related reference document - Do exercises - Self practice	
23	Chapter 07. Client-Server Communication (contd.)	Offline	CLO2	TU	https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction		- Read the slide, related reference document - Do exercises - Self practice	
24	Chapter 08. Fetching and Caching Data	Offline	CLO2	TU	https://react.dev/reference/react/cache		- Read the slide, related reference document - Do exercises - Self practice	
25	Chapter 08. Fetching and Caching Data (contd.)	Offline	CLO2	TU	https://react.dev/reference/react/cache		- Read the slide, related reference document - Do exercises - Self practice	
26	Chapter 08. Fetching and Caching Data (contd.)	Offline	CLO2	TU	https://react.dev/reference/react/cache		- Read the slide, related reference document - Do exercises - Self practice	
27	Chapter 09. React Router and Single Page Application (SPA)	Offline	CLO2	TU	https://www.w3schools.com/react/react_router.asp https://reactrouter.com/en/main/start/tutorial		- Read the slide, related reference document - Do exercises - Self practice	
28	Chapter 09. React Router and Single Page Application (SPA) (contd.)	Offline	CLO2	TU	https://www.w3schools.com/react/react_router.asp https://reactrouter.com/en/main/start/tutorial		- Read the slide, related reference document - Do exercises - Self practice	
29	Chapter 09. React Router and Single Page Application (SPA) (contd.)	Offline	CLO2	TU	https://www.w3schools.com/react/react_router.asp https://reactrouter.com/en/main/start/tutorial		- Read the slide, related reference document - Do exercises - Self practice	
30	Lab 02. Create Single Page Application with ReactJS	Offline	CLO1, CLO2		Lab 02		- Do Lab 02	
31	Lab 02. Create Single Page Application with ReactJS (contd.)	Offline	CLO1, CLO2		Lab 02		- Do Lab 02	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
32	Assignment 01 Guide - Create Single Page Application with ReactJS	Offline	CLO1, CLO2					
33	Project Assistant - 01	Offline		TU	- Discuss project problems		- Discuss project problems	
34	Progress Test 01	Offline		TU			- Take Quiz 01	
35	Chapter 10. RESTful Web Services with Spring Boot	Offline	CLO3	ITU	https://spring.io/guides/tutorials/rest/ https://www.baeldung.com/rest-with-spring-series Spring REST: Building Java Microservices and Cloud Applications: Chapter 01, 03		- Read the slide, related reference documents - Do exercises - Self practice	
36	Chapter 10. RESTful Web Services with Spring Boot (contd.)	Offline	CLO3	TU	https://spring.io/guides/tutorials/rest/ https://www.baeldung.com/rest-with-spring-series Spring REST: Building Java Microservices and Cloud Applications: Chapter 01, 03		- Read the slide, related reference documents - Do exercises - Self practice	
37	Chapter 10. RESTful Web Services with Spring Boot (contd.)	Offline	CLO3	TU	https://spring.io/guides/tutorials/rest/ https://www.baeldung.com/rest-with-spring-series Spring REST: Building Java Microservices and Cloud Applications: Chapter 01, 03		- Read the slide, related reference documents - Do exercises - Self practice	
38	Chapter 10. RESTful Web Services with Spring Boot (contd.)	Offline	CLO3	TU	https://spring.io/guides/tutorials/rest/ https://www.baeldung.com/rest-with-spring-series Spring REST: Building Java Microservices and Cloud Applications: Chapter 01, 03		- Read the slide, related reference documents - Do exercises - Self practice	
39	Chapter 10. RESTful Web Services with Spring Boot (contd.)	Offline	CLO3	TU	https://spring.io/guides/tutorials/rest/ https://www.baeldung.com/rest-with-spring-series Spring REST: Building Java Microservices and Cloud Applications: Chapter 01, 03		- Read the slide, related reference documents - Do exercises - Self practice	
40	Chapter 11. Documenting REST Services	Offline	CLO3	ITU	https://www.baeldung.com/spring-rest-openapi-documentation		- Read the slide, related reference documents - Do exercises - Self practice	
41	Chapter 11. Documenting REST Services (contd.)	Offline	CLO3	ITU	https://www.baeldung.com/spring-rest-openapi-documentation		- Read the slide, related reference documents - Do exercises - Self practice	
42	Chapter 11. Documenting REST Services (contd.)	Offline	CLO3	ITU	https://www.baeldung.com/spring-rest-openapi-documentation		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
43	Chapter 12. Versioning, Paging, Sorting and Testing	Offline	CLO3	ITU	https://bootify.io/spring-boot/pagination-in-spring-boot-rest-api.html Spring REST: Building Java Microservices and Cloud Applications: Chapter 07		- Read the slide, related reference documents - Do exercises - Self practice	
44	Chapter 12. Versioning, Paging, Sorting and Testing (contd.)	Offline	CLO3	ITU	https://bootify.io/spring-boot/pagination-in-spring-boot-rest-api.html Spring REST: Building Java Microservices and Cloud Applications: Chapter 07		- Read the slide, related reference documents - Do exercises - Self practice	
45	Lab 03. RESTful Web Services with Spring Boot	Offline	CLO3	TU	Lab 03		- Do Lab 03	
46	Lab 03. RESTful Web Services with Spring Boot (contd.)	Offline	CLO3	TU	Lab 03		- Do Lab 03	
47	Project Assistant - 02	Offline		TU			- Discuss project problems	
48	Chapter 13. Connecting RESTful Web Service to JPA	Offline	CLO4	ITU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	
49	Chapter 13. Connecting RESTful Web Service to JPA (contd.)	Offline	CLO4	TU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	
50	Chapter 13. Connecting RESTful Web Service to JPA (contd.)	Offline	CLO4	TU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	
51	Chapter 13. Connecting RESTful Web Service to JPA (contd.)	Offline	CLO4	TU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	
52	Chapter 13. Connecting RESTful Web Service to JPA (contd.)	Offline	CLO4	TU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	
53	Chapter 13. Connecting RESTful Web Service to JPA (contd.)	Offline	CLO4	TU	https://spring.io/guides/gs/rest-service		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
54	Lab 04. Spring Boot RESTful Web Service and JPA	Offline	CLO4	TU	Lab 04		- Do Lab 04	
55	Lab 04. Spring Boot RESTful Web Service and JPA (contd.)	Offline	CLO4	TU	Lab 04		- Do Lab 04	
56	Progress Test 02	Offline		TU			- Take Quiz 02	
57	Project Assistant 03	Offline		TU	- Discuss project problems		- Discuss project problems	
58	Chapter 14. Create Single Page Application	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud https://developer.mozilla.org/en-US/docs/Glossary/SPA		- Read the slide, related reference document - Do exercises - Self practice	
59	Chapter 14. Create Single Page Application (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud https://developer.mozilla.org/en-US/docs/Glossary/SPA		- Read the slide, related reference document - Do exercises - Self practice	
60	Chapter 14. Create Single Page Application (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud https://developer.mozilla.org/en-US/docs/Glossary/SPA		- Read the slide, related reference document - Do exercises - Self practice	
61	Chapter 14. Create Single Page Application (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud https://developer.mozilla.org/en-US/docs/Glossary/SPA		- Read the slide, related reference document - Do exercises - Self practice	
62	Chapter 14. Create Single Page Application (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud https://developer.mozilla.org/en-US/docs/Glossary/SPA		- Read the slide, related reference document - Do exercises - Self practice	
63	Lab 05. Create Single Page Application with Spring Boot RESTful API	Offline	CLO4	TU	Lab 05		- Do Lab 05	
64	Lab 05. Create Single Page Application with Spring Boot RESTful API (contd.)	Offline	CLO4	TU	Lab 05		- Do Lab 05	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
65	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
66	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
67	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
68	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
69	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
70	Chapter 15. Implement CRUD Application with RESTful Web Service and ReactJS (contd.)	Offline	CLO4	TU	https://www.baeldung.com/spring-boot-react-crud		- Read the slide, related reference document - Do exercises - Self practice	
71	Assignment 02 Guide	Offline			Assignment 02		- Read the requirements of Assignment 02 - Self practice	
72	Chapter 16. Securing REST Services with Spring Boot	Offline	CLO5	ITU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
73	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
74	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
75	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
76	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
77	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
78	Chapter 16. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	https://www.baeldung.com/spring-boot-api-key-secret https://spring.io/guides/gs/rest-service-cors https://spring.io/guides/tutorials/spring-boot-oauth2/		- Read the slide, related reference documents - Do exercises - Self practice	
79	Lab 06. Securing REST Services with Spring Boot	Offline	CLO5	TU	Lab 06		- Do Lab 06	
80	Lab 06. Securing REST Services with Spring Boot (contd.)	Offline	CLO5	TU	Lab 06		- Do Lab 06	
81	Assignment 03 Guide	Offline	CLO5	TU	Assignment 03		- Read the requirements of Assignment 03 - Self practice	
82	Chapter 17: Hibernate Architecture, Configuration, Mapping	Offline	CLO6	TU	https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
83	Chapter 17: Hibernate Architecture, Configuration, Mapping (contd.)	Offline	CLO6	TU	https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
84	Chapter 17: Hibernate Architecture, Configuration, Mapping (contd.)	Offline	CLO6	TU	https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
85	Chapter 17: Hibernate Architecture, Configuration, Mapping (contd.)	Offline	CLO6	TU	https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
86	Chapter 17: Hibernate Architecture, Configuration, Mapping (contd.)	Offline	CLO6	TU	https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
87	Chapter 18: Hibernate Application Development & CRUD Operations	Offline	CLO6	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
88	Chapter 18: Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO6	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
89	Chapter 18: Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO2	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
90	Chapter 18: Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO6	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
91	Chapter 18: Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO6	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	
92	Chapter 18: Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO6	TU	https://openjfx.io/openjfx-docs/ https://hibernate.org/orm/documentation/6.3/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
93	Lab 7 - Hibernate Application Development & CRUD Operations	Offline	CLO6	TU	Lab 07		- Do Lab 07	
94	Lab 7 - Hibernate Application Development & CRUD Operations (contd.)	Offline	CLO6	TU	Lab 07		- Do Lab 07	
95	Progress Test 03 Assignment 03	Offline	CLO6	TU			Quiz LMS	
96	Project Assistant - 04	Offline	CLO8	TU	- Discuss project problems		- Discuss project problems	
97	Chapter 19. Data Management with MongoDB	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	
98	Chapter 19. Data Management with MongoDB (contd.)	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	
99	Chapter 19. Data Management with MongoDB (contd.)	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	
100	Chapter 20. Spring Data with MongoDB	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	
101	Chapter 20. Spring Data with MongoDB (contd.)	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	
102	Chapter 20. Spring Data with MongoDB (contd.)	Offline	CLO7	TU	https://www.mongodb.com/docs/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 08		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
103	Chapter 21. Build Application with Spring Boot and Spring Data (NoSQL)	Offline	CLO7	TU	https://docs.spring.io/spring-boot/docs/current/reference/html/ https://www.mongodb.com/docs/ https://docs.spring.io/spring-data/mongodb/reference/index.html		- Read the slide, related reference documents - Do exercises - Self practice	
104	Chapter 21. Build Application with Spring Boot and Spring Data (NoSQL) (contd.)	Offline	CLO7	TU	https://docs.spring.io/spring-boot/docs/current/reference/html/ https://www.mongodb.com/docs/ https://docs.spring.io/spring-data/mongodb/reference/index.html		- Read the slide, related reference documents - Do exercises - Self practice	
105	Chapter 21. Build Application with Spring Boot and Spring Data (NoSQL) (contd.)	Offline	CLO7	TU	https://docs.spring.io/spring-boot/docs/current/reference/html/ https://www.mongodb.com/docs/ https://docs.spring.io/spring-data/mongodb/reference/index.html		- Read the slide, related reference documents - Do exercises - Self practice	
106	Lab 08 - Build Application with Spring Boot and Spring Data (NoSQL)	Offline	CLO7	TU	Lab 08		- Do Lab 08	
107	Lab 08 - Build Application with Spring Boot and Spring Data (NoSQL) (contd.)	Offline	CLO7	TU	Lab 08		- Do Lab 08	
108	Project Assistant 05	Offline	CLO8	TU	- Discuss project problems		- Discuss project problems	
109	Chapter 22. Spring Boot - DevOps Tools	Offline	CLO3	IT	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	
110	Chapter 22. Spring Boot - DevOps Tools (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	
111	Chapter 22. Spring Boot - DevOps Tools (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	
112	Chapter 22. Spring Boot - DevOps Tools (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
113	Chapter 22. Spring Boot - DevOps Tools (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	
114	Chapter 22. Spring Boot - DevOps Tools (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-boot/reference/using/devtools.html		- Read the slide, related reference documents - Do exercises - Self practice	
115	Project Evaluation	Offline		TU			- Project demonstration	
116	Project Evaluation (contd.)	Offline		TU			- Project demonstration	
117	Project Evaluation (contd.)	Offline		TU			- Project demonstration	
118	Project Evaluation (contd.)	Offline		TU			- Project demonstration	
119	Project Evaluation (contd.)	Offline		TU			- Project demonstration	
120	Project Evaluation (contd.)	Offline		TU			- Project demonstration	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	3	10.0%	>0	30 slots	CLO1, CLO2, CLO3, CLO4, CLO5			Subjects in syllabus		
2	Group Project	on-going	1	25.0%	>0	38 slots	All CLOs			Subjects in syllabus		
3	Progress test	on-going	3	10.0%	>0	40'	CLO1, CLO2, CLO3, CLO4, CLO5	Multiple choices Marked by Computer or a suitable format	30/each	Subjects in syllabus		
4	Final exam	Final exam	2	55.0%	4	145 minutes	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7,CLO8			All subjects in syllabus	By exam board Manual evaluating	Each part of the final exam must be >=4/10 & Final Result >=5/10.

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
5	Practical Exam	Final exam	1	25.0%	4	85'	CLO2, CLO3, CLO4, CLO5			Subjects in syllabus	By exam board Manual evaluating	
6	Theory Exam	Final exam	1	30.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7		50	All subjects in syllabus	By exam board	